

U.G.C. NET EXAM COMPUTER SCIENCE (Solved with Explanation)

1. The number of positive integers not exceeding 100 that are either odd or the square of an integer is _____ .

(a) 63 (b) 59
(c) 55 (d) 50

Ans. (c) : By using inclusion exclusion principal

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Number of odd number in the range of (1 – 100)
 $= 100/2 = 50$ (1, 3, 5, 7, 97, 99) = $|A|$

Number of squares = 10

$$(1, 4, 9, 16, 25, 36, 49, 81, 100) = |B|$$

Number of odd number and square = 5

$$(1, 9, 25, 49, 81) = |A \cap B|$$

Number of positive integers not exceeding 100 that are either odd or the square = $|A \cup B| = 50 + 10 - 5 = 55$

2. How many ways are there to pack six copies of the same book into four identical boxes. Where a box can contain as many as six books?

(a) 4
(b) 6
(c) 7
(d) 9

Ans. (d) : Here there are six copies of the same book into four.

Identical boxes (Same box). We will enumerate all ways to pack the books for each way to pack the books we will list the number of books in the box with the largest numbers of books, followed by the numbers of books in each box containing at least one book, in order to decreasing the number of books in a box.

6, 0, 0, 0

5, 1, 0, 0

4, 2, 0, 0

4, 1, 1, 0

3, 3, 0, 0

3, 2, 1, 0

3, 1, 1, 1

2, 2, 2, 0

2, 2, 1, 1

So, the number of way is 9.

3. Which of the following pairs of propositions are not logically equivalent?

(a) $(p \rightarrow r) \wedge (q \rightarrow r)$ and $(p \vee q) \rightarrow r$
(b) $p \leftrightarrow q$ and $(\neg p \leftrightarrow \neg q)$
(c) $((p \wedge q) \vee (\neg p \wedge \neg q))$ and $p \leftrightarrow q$
(d) $((p \wedge q \rightarrow r))$ and $((p \rightarrow r) \wedge (q \rightarrow r))$

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Ans. (d) :

Options A:- $(p \rightarrow r) \wedge (q \rightarrow r)$ and $((p \vee q) \rightarrow r)$

Both are equal and it gives the same truth table.

Hence it is logically equivalent.

Options B:- $p \leftrightarrow q$ and $(\neg p \leftrightarrow \neg q)$

$(p \rightarrow q) (q \rightarrow p)$ and $(p \rightarrow q)$

$(\neg q \rightarrow p) (p + \bar{q}) (q + \bar{p})$ and $(p + \bar{q}) (q + \bar{p})$

Both are equal and it gives the same truth table.

Options C:- $((p \wedge q) \vee (\neg p \rightarrow q))$ and $p \leftrightarrow q$

hence logically equivalent.

Options D:- $((p \wedge q \rightarrow r))$ and $((p \rightarrow r) \wedge (q \rightarrow r))$

P	Q	R	$\neg P$	$\neg Q$	$P \Rightarrow Q$	$Q \Rightarrow R$	$(P \Rightarrow R) \cap (Q \Rightarrow R)$
T	T	T	F	F	T	T	T
T	T	F	F	F	F	F	F
T	F	T	F	T	T	T	T
T	F	F	F	T	F	T	F
F	T	T	T	F	T	T	T
F	T	F	T	F	T	F	F
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

The above truth table is not equivalent.

Hence the above statement is true, logically not equivalent.

Hence, the correct answer option d.

4. Consider the following linear programming (LP):

Max, $z = 2x_1 + 3x_2$

Such that $2x_1 + x_2 \leq 4$

$x_1 + 2x_2 \leq 5$

$x_1, x_2 \geq 0$

The optimum value of the LP is

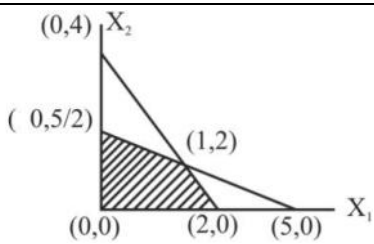
- (a) 23
- (b) 9.5
- (c) 13
- (d) 8

Ans. (d) : Max $z = 2x_1 + 3x_2$

$2x_1 + x_2 \leq 4$

$x_1 + 2x_2 \leq 5$ $x_1, x_2 \geq 0$

Graphical method;



Draw lines by removing in equality z at

$$(1, 2) = 2 + 6 = 8$$

$$z \text{ at } (2, 0) = 4 + 0 = 4$$

$$z \text{ at } (0, 5/2) = 0 + 15/2 = 7.5 = 8 \text{ Ans.}$$

5. Consider a machine with a byte addressable main memory of 2^{16} bytes and block size of 8 bytes. Assume that a direct mapped cache consisting of 32 lines used with this machine. How many bits will be there in Tag, line and word field of format of main memory addresses.
- (a) 8, 5, 3
 (b) 8, 6, 2
 (c) 7, 5, 4
 (d) 7, 6, 3

Ans. (a) : Main memory size = 2^{16} bytes no of blocks in main memory(m) = $2^{16} \div 2^3$
 Block size(p) = 2^3 bytes
 No of block in cache memory (N) = 2^5 bytes
 Tag = 2^8 = Tag bits = $\log_2(M/N)$
 Line or cache block = 2^5 = cache block bits = $\log_2 N$
 word field or word off set 2^3 = word
 offset bits = $\log_2 P$
 Hence, the correct answer is (8, 5, 3).

6. The following program is stored in memory unit of the basic computer. What is the content of the accumulator after the execution of program? (All location numbers listed below are in hexadecimal)

Location	Instruction
210	CLA
211	ADD 217
212	INC
213	STA 217
214	LDA 218
215	CMA
216	AND 217
217	1234 H
218	9CE2H

- (a) 1002H
 (b) 2011H
 (c) 2022H
 (d) 0215H

Ans. (d) :

Location	Instructio	Execution of program
----------	------------	----------------------

	n	
210	CLA	Clean accumulator (set to 0)
211	ADD 217	Add contents of location 217, 1234 H to accumulator
212	INC	Increment accumulator by 1 so accumulator contain 1234 H
213	STA 217	Store this value 1235 H at location 217
214	LDA 218	Load accumulator with the content of 218. So accumulator now has 9CE2H
215	CMA	Complement (1's complement) accumulator, so accumulator now have 631 DH
216	AND 217	perform bitwise AND operation with the content of 217
217	1234 H	
218	9CE2H	
$\Rightarrow (9CE2)_{16} = (1001\ 1100\ 1110\ 0010)_2$ 1's complement of $(9CE2)_{16}$ $= (0110\ 0011\ 0001\ 1101)_2 = (631\ D)_{16}$ $(631\ D)_{16}$ and $(1235)_{16} = (0215)_{16}$ Hence the correct answer is 0215 H		

7. A non-pipeline system takes 50ns to process a task. The same task can be processed in six segment pipeline with a clockcycle of 10ns. Determine approximately the speedup ration of the pipeline for 0 tasks.
- (a) 6

(b) 4.95

(c) 5.7

(d) 5.5

Ans. (b) : Time for non - pipelined execution per task = $t_n = 50\text{ ns}$
time for pipelined execution per task = $t_p = 10\text{ ns}$
number of stages in the pipeline = $K = 6$
number of takes = 500
Calculation :- time for non pipelined = $T_n = 50 \times 500$
Time for pipelined = $T_p = 1 \times 6 \times 10 + (500 - 1) \times 10$
Speed up factor = $S = \frac{T_n}{T_p} = 4.95$

8. What is the radix of the numbers if the solution to the quadratic equation $x^2 - 10x + 26 = 0$ is $x = 4$ and $x = 7$?
- (a) 8

(b) 9

(c) 10

(d) 11

Ans. (d) : If one solve the quadratic equation,

$ax^2 + bx + c$, then the sum of root of quadrate equation = $-\frac{b}{a}$

Quadratic equation $x^2 - 10x + 26 = 0$

Since 4 and 7 are the root of the equation

$$-\frac{b}{a} = \frac{(10)r}{(1)r} = (4)r + (7)r$$

$$r = 4 + 7 = 11$$

9. Consider the following pseudo-code fragment, where a and b are integer variables that have been initialized.

/*Pre-conditions : $(a > 1 \wedge a < b)$ */

/*Assume that overflow never occurs */

int x = 0; int p = 1;

while (p < b) {

p = p * a;

x = x + 1;

}

When the while loop terminates, what will be the value of x is terms of a and b ?

- (a) a^b
- (b) b^a
- (c) $\left\lfloor \log_a^b \right\rfloor$ /* $\lfloor \rfloor$ means floor */
- (d) $\left\lceil \log_a^b \right\rceil$ /* $\lceil \rfloor$ means ceil */

Ans. (d) : Since $a > 1$ and $a < b$

int x = 0, int p = 1

Let us assume q = 2 and b = 15

Whenever the 100 p is entered the value of P is.

doubles and the value of x incremented by 1.

in iteration 1 : P = 2, x = 1

in iteration 2 : P = 4, x = 2

in iteration 3 : P = 8, x = 3

in iteration 4 : P = 16, x = 4

final values are a = 2, b = 15, p = 16, x = 4.

Form the above value if can be seen that $x = \left\lceil \log_a^b \right\rceil$

10. In HTML <map> tag is used for

- (a) defining a path between two nodes in an image
- (b) defining clickable region in an image
- (c) highlighting an area in an image
- (d) defining the site amp of a web-site

Ans. (b) :

• The <map> tag is used to define an image map. An Image map is an image with clickable areas.

• The required name attribute of the <map> element is associated with the 's use map attribute and creates a relationship b/w the image and the map.

- An image map is made up of an image with clickable areas, When you click on the image it will open to a new or previously selected destination.

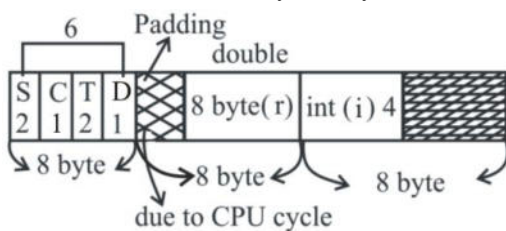
11. Suppose you are compiling on machine with 1-byte chars, 2-byt shorts, 4-byte ints and 8-byte doubles and with alignment rules that require the address of every primitive data element to be an integer multiple of the element's size. Suppose further that the compiler is not permitted to reorder fields padding is used to ensure alignment. How much space will be consumed by the following array.

```
struct {
    short s;
    char c;
    short t;
    char d;
    double r;
    int i;
} A[10]; /*10-element array of structs*/.
```

- (a) 150 bytes (b) 320 bytes
(c) 240 bytes (d) 200 bytes

Ans. (c) : Padding it save are CPU cycle.

given data Assume: → CPU cycle 8 byte



$$\Rightarrow 8 \times 3 = 24$$

Array of size = 10 $\Rightarrow 24 \times 10 = 240$ Ans

Data type	32 - bit(4 bytes)	64 - bit(8 bytes)
Char	1	1
Short	2	2
Int	4	4
Floot	4	4
Double	8	8
Pointer	4	8

12. Consider the following recursive Java function f that takes two long arguments and returns a float value :

```
public static float f (long m, long n) {
    float result = (float)m /(float)m;
    if (m < 0 || n < 0) return 0,0f;
    else result = f (m * 2, n * 3);
    return result;
}
```

Which of the following real values best approximates the value of $f(1, 3)$?

- (a) 0.2
- (b) 0.4
- (c) 0.6
- (d) 0.8

Ans. (a) : The function $f(1, 3)$ pass the value to m, n then the value will type cast to float and result = 0.3333 (1/3).

Then it enters into else statement the function result as 0.2 (2/9).

Hence, the correct answer is 0.2.

13. The data node and name node in HADOOP are

- (a) Worker Node and Master Node respectively
- (b) Master Node and Worker Node respectively
- (c) Both Worker Nodes
- (d) Both Master Nodes

Ans. (a) : The main difference between Name Node and data node in HADOOP is that the Name Node is master node in HADDOP distributed file system (HDFS) that manages the file system meta data distributed file system that stores the actual data as instructed by the Name Node.

14. Consider a relational scheme $S = (U, V, W, X, Y, Z)$ on which the following functional dependencies hold.

$\{U \rightarrow V, VW \rightarrow X, Y \rightarrow W, X \rightarrow U\}$

Which are the candidate keys among following options ?

- (a) UY, VY
- (b) UY, VY, XY
- (c) UYZ, VYZ, VWZ
- (d) UYZ, VYZ, XYZ

Ans. (d) : A candidate key is a column or set of columns in a table that can uniquely identify any data base record without referring to any other data.

The following functional dependencies,

$U \rightarrow V, VW \rightarrow X, Y \rightarrow W, X \rightarrow U$

$S = (U, V, W, X, Y, Z)$

Here YZ an independent key, so for every key, the YZ Should their. Hence only option 4 is containing YZ for all keys. Hence the correct answer is UYZ, VYZ, XYZ.

15. Consider a hypothetical machine with 3 pages of physical memory, 5 pages of virtual memory, and $\langle A, B, C, D, A, B, E, A, B, C, D, E, B, A, B \rangle$ as the stream of page references by an application. If P and Q are the number of page faults that the application would incur with FIFO and page replacement algorithms respectively, then $(P, Q) = \underline{\hspace{1cm}}$. (Assuming enough space for storing 3 page frames)

- (a) (11, 10)
- (b) (12, 11)
- (c) (10, 11)
- (d) (11, 12)

Ans. (d) :

• **Key point :-** 3 pages of physical memory mean 3 frames (So at a time only max 3 pages in frames) 5 pages of virtual memory mean 5 page.

• **FIFO (First in, first out):-** The operating system uses this algorithm to keep track of all pages in memory in a queue with the oldest page at the top. When a page has to be replace the first page in the queue is chosen for replacement.

Page faults - ABC DABF CDBA

A	D	D	D	E	E	E	B	B
B	B	A	A	A	C	C	C	A
C	C	C	B	B	B	D	D	D

Page Fault = 11

Least Recently used (LRU):- In this algorithm the page will be replace which is the least recently used. Page

fault = ABCDABECDEBA

A	D	D	D	E	C	C	C	B	B
B	B	A	A	A	A	D	D	D	A
C	C	C	B	B	B	B	E	E	E

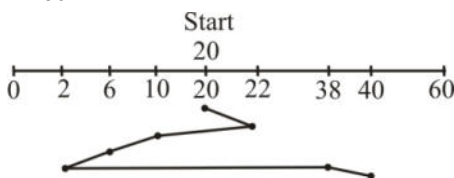
Page fault = 12

16. Consider a disk system having 60 cylinders, Disk requests are received by a disk drive for cylinders, 10, 22, 20, 2, 40, 6 and 38 in the order. Assuming the disk head is currently at cylinder 20, what is the time taken to satisfy all the requests if it takes milliseconds to move from one cylinder to adjacent one and Shortest Seek Time First (SSTF) algorithm is used ?

- (a) 240 milliseconds
- (b) 96 milliseconds
- (c) 120 milliseconds
- (d) 112 milliseconds

Ans. (c) : Shortest Seek Time First (SSTF) :-

$$\begin{aligned}
 \text{SSTF} &= 22-20+22-10+10-6+6-2+38-2+40-38 \\
 &= 2 + 12 + 4 + 4 + 36 + 2 \\
 &= 60
 \end{aligned}$$



It takes 2 milliseconds to move from one cylinder to adjacent one = 60×2

$$= 120 \text{ milliseconds}$$

17. Suppose you have a Linux file system where the block size is 2K bytes, a disk address is 32 bits and in i-node contains the disk addresses of first 12 direct blocks of file, a single indirect block, and a double indirect block.

Approximately, what is the largest file that can be represented by an i-node?

- (a) 513 Kbytes
- (b) 513 MBytes
- (c) 537 Mbytes
- (d) 537 KBytes

Ans. (b) : Size of the disk block = 2048 bytes
disk address = 32 bits = 4 bytes

Number of addresses per block = $\frac{2048}{4} = 512 = 2^9$

Maximum file size = 12 direct blocks + 1 indirect
Block + 1 double indirect block

$$= 12 + 2^9 + 2^9 * 2^9$$
$$= 12 + 2^9 + 2^{18}$$

the size of each block = 2^{11} bytes

Maximum size of the file = $(12 + 2^9 + 2^{18}) \times 2^{11}$ bytes

$\therefore$ Maximum size of the file = 5130.236 $\cong$ 513 MBytes

- 18. Consider a single-level page table system, with the page table stored in the memory. If the hit rate to TLB is 80%, and it takes 15 nanoseconds to search the TLB and 150 nanoseconds to access the main memory, then what is the effective memory access time, in nanosecond?**

- (a) 150
- (b) 195
- (c) 205
- (d) 175

Ans. (b) : TLB hit ratio = $p = 0.8$

TLB access time = 15 nano seconds

memory access time = $m = 150$ milliseconds

Formula : $EMAT = P \times (t+m) + (1-p) \times (t+m+n)$

$EMAT = 0.8 \times (15 + 150) + (1 - 0.8) \times (15 + 150 + 150)$

$EMAT = 195$ milliseconds.

- 19. Which of the following UML diagrams has static view ?**

- (a) Collaboration diagram
- (b) Use-Case diagram
- (c) State chart diagram
- (d) Activity diagram

Ans. (b) : A use case diagram only depicts the system's features while a dynamic model view depicts all functions and behaviour.

• A use case diagram is used to represent the dynamic behaviour of a system. It encapsulates the system's functionality by incorporating use cases, actors and their relationships.

- 20. Modifying the software by restructuring is called**

- (a) Adaptive maintenance
- (b) Corrective maintenance
- (c) Perfective maintenance

(d) Preventive maintenance

Ans. (c) : Corrective, adaptive, perfective and preventive maintenance are the four forms of maintenance.

- Corrective maintenance focuses on resolving issues that arises when the programme is in operation.
- Adaptive maintenance is concerned with the changes made to the software in order to make it adaptable to new. Environments such a running it on a new operating system.
- The term "perfective maintenance" refers to the change in the software that occur as a result of adding new features.
- Preventive maintenance entails making adjustments to avoid mistakes from occurring

- 21. A company has a choice of two languages L_1 to develop a software for their client. Number of LOC required to develop an application in L_2 is thrice the LOC in language L_1 . are given below to decide which language should be preferred for development.**

PERAMETER	L_1	L_2
Man year needed for development	LOC/1000	LOC/1000
Development	Rs. 70000	Rs. 90000
Cost of Maintenance per year	Rs. 100000	Rs. 40000

Total cost of project include cost of development and maintenance. What is the LOC for L_1 for which cost of developing the software with both language must be same.

- (a) 2000 (b) 6000
(c) 3000 (d) 5000

Ans. (c) : The total cost of the project = Total development + total maintenance cost

Sol. Let P_1 be the LOC using L_1 and P_2 be the LOC using L_2 .

Total cost of the project using $L_1 = (P_1/1000) \times 70,000 + (10 \times 100000) = 70 P_1 + 100000$

Total cost of the project using $L_2 =$
 $(P_2/1000) \times 90000 + (10 \times 40000)$
 $= 90P_2 + 400000$

Given that $P_2 = 3 \times P_1$

$$\begin{aligned} \Rightarrow 70P_1 + 1000000 &= 90P_2 + 400000 \\ \Rightarrow 70P_1 + 1000000 &= 270P_1 + 400000 \\ \Rightarrow 200P_1 &= 600000 \\ \Rightarrow P_1 &= 3000 \end{aligned}$$

- 22. A Software project was estimated at 864 Function Point. A six person team will be assigned to project consisting of a requirement gathering person, one designer, two programmers and two testers. The salary of the**

designer is ₹ 70000 per month, requirement gatherer is ₹ 50000 per month programmer is ₹ 60000 per month and a tester is ₹ 60000 per month. Average productivity for the team is 12 FP per person month. Which if the following represents the projected cost of the project?

- (a) ₹ 3320000
- (b) ₹ 4320000
- (c) ₹ 3310000
- (d) ₹ 2210000

Ans. (b) : given data functional point = 864

team = 6 persons

Average each person (FP) productivity = 12

designer month salary = 70000

Programmers month salary = 2×60000

tester month salary = 2×60000

Requirement gatherer = 50000

Projected cost of project = ?

total number of months

$$= \frac{\text{functional point (FP)}}{\text{Team} \times \text{average each person (FP) productivity}}$$

$$\frac{864}{6 \times 12} = 12 \text{ months}$$

Total cost of the project = (one designer salary + $2 \times$ programmer month salary + $2 \times$ tester month salary + one requirement gatherer) $\times$ total number of months

$$= (70000 + 2 \times 60000 + 2 \times 60000 + 50000) \times 12$$

$$= 4320000$$

23. A complete 3-ary tree is a tree in which each node has n children or no children. Let I be the number of Internal nodes and L be the number of leaves in a complete n-ray tree. If L = 41 and I = 10. What is the value of n ?

- (a) 3
- (b) 4
- (c) 5
- (d) 6

Ans. (c) : If the tree is 3-ary and 'l' is an internal node the number of leaves $2l + 1$.

• If the tree is 4-ary and 'l' an internal node, the number of leaves is $4l + 1$.

of the tree is n-ary and 'l' is an internal node the number of leaves is $(n - 1)l + 1$.

given that leaves $L = 41$, internal nodes $d = 10$

$$L = (n - 1)l + 1$$

$$41 = 10(n - 1) + 1$$

$$10n = 50$$

$$n = 5$$

24. In a binary max heap containing a number, the smallest element can be found in _____ time.

- (a) $O(n)$
- (b) $O(\log_2 n)$

- (c) $O(1)$
 (d) $O(\log_2 \log_2 n)$

Ans. (a) : The heap's smallest values will always be present on the last level or second last of the heap in a max heap and the time complexity of reaching the last level of the Heap is $O(n)$.

- It will take $O(n)$ time to traverse all of the elements using linear search.

25. If algorithm A and another algorithm B take $\log_2(n)$ and $\sqrt{n}$ microseconds, respectively, to solve a problem, then the largest size n of a problem these algorithm can solve, respectively, in one second are ____ and ____ .

- (a) 2^{10^6} and 10^6
 (b) 2^{10^6} and 10^{12}
 (c) 2^{10^6} and $6 \cdot 10^6$
 (d) 2^{10^6} and $6 \cdot 10^{12}$

Ans. (b) : A microsecond is 10^{-6} seconds.

Hence one seconds = 10^6 microsecond

One hour = $60 \times 60 \times 10^6 = 3.6 \times 10^9$ microseconds

One month = 2.592×10^{12} microseconds

One century = 3.1104×10^{15} microseconds

$f(n) = \log n$ in this case the largest value n

such that $\log n \leq 10^6$

We rewrite as $2^{\log n} \leq 2^{(10) \wedge 6}$

thus, $2^{1000000} = 2^{(10)100000} = (10^3)^{100000} = 10^{300000}$

Hence $\log_2(n)$ and $\sqrt{n}$ in one seconds 2^{10^6} and 10^{12} .

26. Let G be a directed graph whose vertex set is the set of numbers from 1 to 100. There is an edge from a vertex i to a vertex j if and only if either $j = i + 1$ or $j = 3i$. the minimum number of edges in a path in G from vertex 1 to vertex 100 is ____ .

- (a) 23 (b) 99
 (c) 4 (d) 7

Ans. (d) : Edge set consists of edges from i to $0j$, using either two conditions are $j = i + 1$ or $j = 3i$

Second choice helps us to move from 1 to 100. The trick to slot this is to think the other way around. Try to find a 100 to 1 trail instead of having a 1 trail to 100.

So, the edge sequence with the minimum number of edges is - $1 \rightarrow 3 \rightarrow 9 \rightarrow 10 \rightarrow 11 \rightarrow 33 \rightarrow 99 \rightarrow 100$

Which consists of 7 edges.

27. Consider $L = L_1 \cap L_2$

Where $L_2 = \{0^m 0^n 20^n 1^n \mid m, n \geq 0\}$

$L_1 = \{0^m 1^n 2^k \mid m, n, k \geq 0\}$

Then, the language L is

- (a) Recursively enumerable but not context free
 (b) Regular
 (c) Context free but not regular

(d) Not recursive

Ans. (c) : L_1 will first contain 0's followed by an equal number of 1's. After that there will be a single 2. Which will again be followed by 0's and an equal number of 1's i.e.

$L_1 = \{2, 012, 00112, 20011, 01201, 0011201, 012001\}$

$L_2 = \{\text{Epsilon}, 0, 1, 2, 01, 12, 012, 00112, 20011, 01201, 0011201, 0120011, 001120011, \dots\}$

$L = L_1 \cap L_2 = \{2, 012, 00112, 0001112, \dots\}$

$L = L_1 \cap L_2 = \{0^m 1^{m/2} \mid m \geq 0\}$, Which is a context free language but not regular language.

28. Let L_1 and L_2 be language over $\Sigma = \{a, b\}$ represented by the regular expressions $(a^* + b)^*$ and $(a + b)^*$ respectively.

Which of the following is true with respect to the two languages?

(a) $L_1 \subset L_2$

(b) $L_2 \subset L_1$

(c) $L_1 = L_2$

(d) $L_1 \cap L_2 = \phi$

Ans. (c) : $(a + b)^*$ is a universal set Grammar which can generate all strings over all possible alphabet "a" and "b"

$(a^* + b)^* = a^*$ can generate epsilon and remaining part is like

$(a + b)^*$

$(a^* + b)^* = (a + b^*)^* = (a^* b^*)^* = (a^* + b^*)^*$

$= a^* (ba^*)^* = b^* (ab^*)^* = (a + b)^*$

all grammars are equal-

Hence the correct answer is $L_1 = L_2$

29. Which of the following statements is true ?

(a) The union of two context free languages is context free.

(b) The intersection of two context free language is context free.

(c) The complement of a context free language is context free.

(d) If a language is context free, it can always be accepted by a deterministic pushdown automaton.

Ans. (a) : Context free language is closed under, union concatenation and Kleen closure. But not closed under complementation and intersection.

• If a language is context free it can be either deterministic or non-deterministic language only deterministic language are accepted by the deterministic pushdown automation.

30. Which of the following is not an intermediate code form?

(a) Syntax trees

(b) Three address codes

(c) Quadruples

(d) Post fix Notation

Ans. (c) : Intermediate code can be represented in three forms, which are post fix notation syntax tree, three address code.

- In a compiler three address code can be implemented as records with field for operator and operands. There are three such representations, quadruples, triples indirect triples.

31. Consider a code with only four valid code words 0000000000, 0000011111, 1111100000, and 1111111111. This code has distance 5. If the code word arrived is 0000000111 then the original code word must be ____ .

- (a) 0000011111
- (b) 0000000000
- (c) 1111100000
- (d) 1111111111

Ans. (a) : Arrange one below the other the XOR operations.

for correcting the d-bit error hamming distance should be $2d + 1$.

So, $2d + 1 = 5$

$d = 2$. It means the hamming distance b/w the code word 0000000111 and given options should be 2-bits only options 1 match.

32. Using 'RSA' public key cryptosystem. if $P = 3$, $q = 11$ and $d = 7$, find the value of a and energy the number '19'.

- (a) 20, 19
- (b) 33, 11
- (c) 3, 25
- (d) 77, 28

Ans. (c) : $P = 3$, $q = 11$ and $d = 1$

$$\phi(n) = (p - 1) \times (q - 1)$$

$$n = p \times q$$

$$n = 3 \times 11 = 33$$

$$\phi(n) = (3 - 1) \times (11 - 1) = 20$$

$$e \times 7 \bmod \phi(n) = 1$$

Option -3

$$\text{if } e = 3, 7 \times 3 \bmod 20 = 1$$

Encrypt key $M^e \bmod n = c$

$$c = 19^3 \bmod 33$$

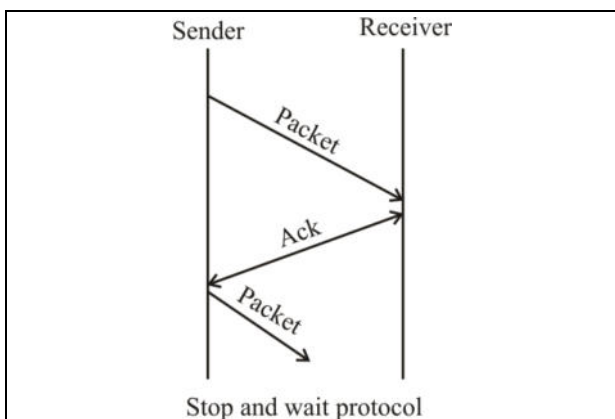
$$c = 28$$

$\therefore$ Hence answer is (3, 28)

33. Protocols in which the sender sends one frame and then waits for an acknowledgment before proceeding for next frame are called as ____ .

- (a) Simplex protocols
- (b) Unrestricted simplex protocols
- (c) Simplex stop and wait protocols
- (d) Restricted simplex protocols

Ans. (c) :



Sender side -

Rule-1 Sender sends one data packet at a time.

Rule-2 Sender sends the next packet only when it receives the acknowledgment of the pervious packet.

Receiver side -

Rules- 1 Receive and then consume the data packet.

Rule-2 When the data packet is consumed receiver sends the acknowledgment to the sender.

- The idea of stop and wait protocol in the receivers side is also very simple consumed the acknowledgment is sent this know as a flow control mechanism.

34. The period of a signal is 100ms. Then the frequency of this signal in kilohertz is ____ .

- (a) 10 (b) 10^{-1}
(c) 10^{-2} (d) 10^{-3}

Ans. (c) : period = T = 100 ms

$$\text{Frequency} = \frac{1}{T}$$

$$f = \frac{1}{100} = \frac{1}{100 \times 10^{-3} \text{s}} = 10^{-1} \text{s} = 10 \text{ H}$$

$$f = 10 \times 10^{-3} \text{ kHz} = 10^{-2} \text{ kHz}$$

35. The process of removing details from a given state representation is called ____ .

- (a) Extraction
(b) Mining
(c) Selection
(d) Abstraction

Ans. (d) : The "process of removing detail" from a given state representation is called an abstraction.

- Abstraction is a type artificial Intelligence problem formulation and problem solving process.

36. Which of the following is NOT true in problem solving in artificial intelligence ?

- (a) Implements heuristic search technique
(b) Solutions steps are not explicit
(c) Knowledge is imprecise
(d) It works on or implement repetitions mechanism

Ans. (d) : Every all program has to do the process of searching for the solution steps that are not explicit in nature.

- Knowledge plays a vital role in intelligence as well as creating artificial intelligence.
- Hence the incorrect statements. it work on or implements a repetition mechanism.

37. If $f(x) = x$ is my friend and $p(x) = x$ i as perfect, then correct logical translation of the statement "some of my friends are not perfect" is ____ .

- (a) $\forall_x(f(x) \wedge \neg p(x))$
- (b) $\exists_x(f(x) \wedge \neg p(x))$
- (c) $\neg(f(x) \wedge \neg p(x))$
- (d) $\exists_x(\neg f(x) \wedge \neg p(x))$

Ans. (b) : $f(x) = x$ is my friend

$p(x) = x$ i as perfect,

So, that are asking about some, finally outer most parentheses will get.

SOME $\exists x$.

They one given conditions like NOT perfect.

So, we get $P(x)$ the final condition is $\exists_x(f(x) \wedge \neg p(x))$.

38. What king of clauses are available in conjunctive normal form ?

- (a) Disjunction of literals
- (b) Disjunction of variables
- (c) Conjunction of literals
- (d) Conjunctions of variables

Ans. (a) : In Boolean logic conjunctive normal or CNF means the product of sums (POS) which clearly states that is a product of clauses which in turn are the sum of literals.

Ex. $y' = (A + B)(A' + B)$

y is in the form of conjunctive normal form it is a product of $(A + B)$ and $(A' + B)$ which are sums of A and B , A' and B .

39. Consider the following properties:

A. Reflexive

B. Antisymmetric

C. Symmetric

Let $A = \{a, b, c, d, e, f, g\}$ and $R = \{(a, a), (b, b), (c, d), (c, g), (d, g), (e, e), (f, f), (g, g)\}$ be a relation of A. Which of the following property (properties) is (are) satisfied by the relation R?

- (a) Only A
- (b) Only C
- (c) Both A and B
- (d) B and Not A

Ans. (d) : If a binary relation R over a set X relates every element of x to itself. It is said to be reflexive since (c. e) and (a, d) is not given in the given question it is not reflexive.

- A binary relation is anti symmetric is there is no pair of distinct elements of X each of which is related by R to the other. The given relation R is anti symmetric

because for (c, d), (d, c) is not present in R. Similarly for (c, g) and (d, g).

A binary relation is a type of binary relation. An example is the relation is equal to because if $a = b$ is true then $b = a$ is also true.

Here since (c, d) is pair in a given relations R for which (d, c) is not present in it. So it violates the symmetric property of the relation, Hence it is not symmetric.

40. Consider the following argument with premise $\forall x(P(x) \vee Q(x))$ and conclusion $(\forall x P(x)) \wedge (\forall x Q(x))$

(A)	$\forall x(p(x) \vee Q(x))$	Premise
(B)	$P(c) \vee Q(c)$	Universal instantiation from (A)
(C)	$P(c)$	simplification from (B)
(D)	$\forall x P(x)$	Universal Generalization of (C)
(E)	$Q(c)$	Simplification from (B)
(F)	$\forall x Q(x)$	Universal Generalization of (E)
(G)	$(\forall x P(x)) \wedge (\forall x Q(x))$	Conjunction of (D) and (F)

- (a) This is a valid argument
 (b) Steps (C) and (E) are not correct inferences
 (c) Steps (D) and (F) are not correct inferences
 (d) Step (G) is not a correct inference

Ans. (b) : For C and E to be true should be $P(c) \cap Q(c)$. Hence, this is not a correct inference.

- In predicate logic universal generalization states that if ... $p(x)$ has been derived then... $\forall x P(x)$ can be derived.

41. Which of the following are applications of symbol table ?

- (A) Storage allocation
 (B) Checking type compatibility
 (C) Suppressing duplicate error messages

Choose the correct answer from the option given below

- (a) (A) and (B) only
 (b) (A) and (C) only
 (c) (B) and (C) only
 (d) (A), (B) and (C)

Ans. (d) : Storage allocation -

- Checking type compatibility.
- Suppressing duplicate error messages.

The symbol table is a data structure that contains a record for each identifier, along with its attribute such as type, initial value scope a lifetime.

42. Simplified expression's for following Boolean function
 $F(A, B, C, D) = \sum(0, 1, 2, 3, 6, 12, 13, 14, 15)$ is/are
- (A) $A'B' + AB A'C'D'$
 - (B) $A'B' + AB + A' CD'$
 - (C) $A'B' + AB + BC'D'$
 - (D) $A'B' + AB + BCD'$
- Choose the correct answer the from the option's given below.
- (a) (A) only
 - (b) (B) only
 - (c) (A) and (B) only
 - (d) (B) and (D) only

Ans. (d) :

$F(A, B, C, D) = \sum(0, 1, 2, 3, 6, 12, 13, 14, 15)$

Two k-maps can be constructed from the given Boolean function.

$AB \backslash CD$	00	01	11	10	
00	1 0	1 1	1 3	1 2	→ $A'B'$
01				1 6	→ $A'CD'$
11	1 12	1 13	1 15	1 14	→ AB
10					

$K=Map^1$

$AB \backslash CD$	00	01	11	10	
00	1 0	1 1	1 3	1 2	→ $A'B'$
01				1 6	→ BCD'
11	1 12	1 13	1 15	1 14	→ AB
10					

$K=Map^2$

The expression for k-map is $AB + A'B' + A' CD'$.

The expression for k-map is $AB + A'B' + BCD'$

43. Which of the following statements with respect to K-segment pipelining are true ?
- (A) Maximum speedup that a pipeline can provide is k theoretically.
 - (B) It is impossible to achieve maximum speedup k in k-segment pipeline.
 - (C) All segments in pipeline take same time in computation.
- Choose the correct answer from the options given below:
- (a) (A) and (B) only
 - (b) (B) and (C) only
 - (c) (A) and (C) only
 - (d) (A), (B) and (C)

Ans. (a) : Pipelining :- overlapping execution - Parallelism improve performance. The aim of using pipelining is to use a single clock per instruction (CPI) which automatically improves performance.

Statement A is true-

$S = \frac{t_n}{t_p}$ if we assume that the time it takes to process a task is the same in the pipeline and pipelined circuits will have $t_n = kt_p$.

Including this assumption the speedup reduces to

$$S = \frac{kt_p}{t_p} = k$$

Statements B true- $k \times \text{efficiency} = \text{speedup}$

The speedup cannot be equal to k till efficiency become 1 and in real time environment efficient cannot become 1 of many reasons like the delay at intermediate buffer different time taken by different segments to perform their sub-operations which cause all other segments to waste time while waiting for the next clock.

Statement C False : Because different segment can take different times to complete their sub operations. Hence the correct answer is (A) and (B) only.

44. Which of the following statements with respect to multiprocessor system are true ?

- (A) Multiprocessor system is controlled by one operating system.
- (B) In Multiprocessor system multiple computers are connected by the means of communications lines.
- (C) Multiprocessor system is classified as multiple instruction stream and multiple data stream system.

Choose the correct answer from the options given below:

- (a) (A) only
- (b) (A) and (B) only
- (c) (A) and (C) only
- (d) (B) and (C) only

Ans. (c) : A multiprocessor is a computer System having two or more processing units. multiple processors each Sharing main memory and Peripherals in order to Simultaneously Process programs.

- multiprocessor System are Controlled by are or more operating systems. Hence Statements I is correct.

- in a multiprocessor system, multiple Processors are Connected Under single Computer System. Hence Statement II is incorrect.

- Multiprocessor System is classified us multiple Instruction streams and multiple data streams system. Hence statement III is correct.

$\Rightarrow$ Correct option is c (A) and (C) only

45. Which of the following statements regarding XML is/are True?

- (A) XML is a set of tags designed to tell browsers how to display text and images in a web page.**
- (B) XML defines a syntax, for representing data, but the meaning of data varies from application, to application.**
- (C) < Letter>, < LETTER > and < letter > are three different tags is XML.**

Choose the correct answer from the options given below:

- (a) (A) and (B) only (b) (A) and (C) only**
- (c) (B) and (C) only (d) (A), (B) and (C)**

Ans. (c) : • XML was created with the aim of transporting data with an emphasis on what data is.
• HTML was developed with the intention of presenting data with an emphasis on how it appears.
Statement B: TRUE:-
While XML Specifies a syntax for representing the meaning of data varies by application.
Statements C : TRUE
XML is case sensitive and Hence <Letter> , <LETTER> and < letter> are three different tags in XML
Statement :- A: FALSE:
HTML is a collection of tags that tell web browsers How to view text and Images on website.

46. In the context of 3D computer graphics which of the following statements is/are correct?

- (A) Under perspective, projection, each set of parallel lines in the object do not stay parallel in the image (except those that are parallel to the viewplane to start with).**
- (B) Applying a perspective, transformation in the graphics pipeline to a vertex involves dividing by its 'z' coordinate.**
- (C) Perspective transformation is a linear transformation.**

Choose the correct answer form the options given below:

- (a) (A) and (B) only (b) (A) and (C) only**
- (c) (B) and (C) only (d) (A), (B) and (C)**

Ans. (a) : Under perspective projection, Each set of Parallel Lines in the object do not stay parallel in the image. The Lines Converge to the middle of the Projection at a single points. points of Intersection of Converging line with the screen plane generate the projected Image on the Screen.
Hence statement I is correct.
• Appling a perspective transformation in the graphics Pipeline to a vertex involves dividing by its' 'z' Coordinate.
Hence statement II is correct.
• perspective transformation is a non- Linear transformation.
Hence statement III is incorrect.

47. Concerning phong shading and Gouraud shading in a 3D scene, which of the following statements are true?

(A) Gouraud shading requires more computation than phong shading.

(B) Gouraud shading linearly interpolates the color of an interior pixel from the color at the vertices.

(C) Phong shading interpolates over the normal vectors specified at the vertices.

Choose the correct answer from the options given below:

(a) (A) and (B) only (b) (A) and (C) only

(c) (B) and (C) only (d) (A), (B) and (C)

Ans. (c) : Statement 1: FALSE:-

Gouraud Shading requires less calculation and this greatly decreases the cost of shading steeply.

Statement - 3 : True :- Phong Shading Interpolates over the normal vector specified at the vertices.

Statement - 2 :- TRUE :-

Gouraud Shading Linearly interpolates the color of an interior pixel from the color at the vertices the original is a linear interpolation of Colour between vertices, Specifically.

48. In the context of concurrency control, a given pair of operations in a schedule is called conflict schedule if

(A) At least one of the operations is write operation.

(B) Both the operations are performed on the same data item.

(C) Both the operations are performed by different transactions.

(D) Both the operations are performed on different data items.

Choose the correct answer from the options given below:

(a) (A) and (B) only (b)(A), (B) and (C) only

(c) (A), (C) and (D) only (d) (C) and (D) only

Ans. (b):

(a) item X has an incorrect value because its update by T_1 is lost (over written)

T_1	T_2
read_item(X); $X = X - N$; Write_item(X); read_item(Y) $Y = Y + N$; write_item(Y);	read_item(X); $X_2 = X + m$; Write_item(X);

(b) Transaction T_1 fails and must change the value of x back to its value, meanwhile, T_2 has read the temporary incorrect value of x .

T_1	T_2
$\text{read_item}(X);$ $X = X - N;$ $\text{Write_item}(X);$ $\text{read_item}(Y)$	$\text{read_item}(X);$ $X = X + m;$ $\text{Write_item}(X);$

T_3 reads X after N is Subtracted and reads Y before N is added a wrong summary is a result (off by N)

T_1	T_3
$\text{read_item}(X);$ $X = X - N;$ $\text{Write_item}(X);$ $\text{read_item}(Y)$ $Y = Y + N;$ $\text{write_item}(Y);$	$\text{SUM} = 0$ $\text{redd_item}(A)$ $\text{Sum} = \text{Sum} + A;$ $\text{read_item}(X)$ $\text{Sum} = \text{Sum} + X$ $\text{read} = \text{item}(Y);$ $\text{SUM} = \text{SUM} + Y;$

Two function are said to be incompatible if only of the following Conditions are met.

- They are associated with various Transactions.
- They both work with the same data object.
- One of them must be write operation.
- If you note the issues mentioned below, it's because different transactions have performed at least one write operation on the same data object.

49. Software reliability is described with respect to
(A) Execution Time
(B) Calendar Time
(C) Clock Time
Choose the correct answer from the options given below:

- (a) (A) and (B) only (b) (B) and (C) only
(c) (A), (B) and (C) (d) (A) and (C) only

Ans. (c): The Execution time of a programme is the time that a processor spends actually executing the programs instructions.

- The second kind of time is calendar time it is the familiar time that we normally experience.
- Clock time is the normal wall clock time.
- and software reliability is described with respect to all these.

50. To create an object-behavioral model, the analyst performs the following steps:
- (A) Evaluates all use-cases.
 - (B) Builds state transition diagram for the system.
 - (C) Reviews the object behaviour model to verify accuracy and consistency.
 - (D) Identifies events that do not derive the interactions sequence.

Choose the correct answer from the options given below:

- (a) (A), (B) and (C) only
- (b) (A), (B) and (D) only
- (c) (B), (C) and (D) only
- (d) (A), (C) and (D) only

Ans. (a): The behavioral model indicates how the Software will respond to external events. To create the model, the analyst must perform the following steps

1. Evaluate all use - cases to fully understand the sequence of interaction within the system
2. Create a sequence for each use - case
3. Build a state diagram for the system.
4. Review the behavioral model to verify accuracy and Consistency.

51. Which of the following is/are behavioral testing technique(s) ?
- (A) Equivalence partitioning
 - (B) Graph Based Testing Method
 - (C) Boundary Value Analysis
 - (D) Data Flow Testing
 - (E) Loop Testing

Choose the correct answer from the options given below:

- (a) (B) and (D) only
- (b) (A), (B) and (C) only
- (c) (D) and (E) only
- (d) (A), (C) and (E) only

Ans. (b) : Black- box testing, also known as behavioral testing, focuses on the software's, functional specifications. In other words, black - box testing allows a software developer to create sets of input conditions that completely exercise all of a program's functional requirements. The methods for behavioral testing are as follows.

- (1) graph -based testing methods
- (2) Equivalence partitioning
- (3) boundary value analysis
- (4) Comparison testing
- (5) orthogonal array testing

52. The running time of an algorithm is $O(g(n))$ if and only if
- (A) its worst-case running time is $O(g(x))$ and its best case running time is $\Omega(g(n))$. ($\theta = \text{big } \theta$)
 - (B) its worst-case running time is $\Omega(g(n))$ and its best-case running time is $\theta(g(n))$. ($\theta = \text{big } \theta$)

(C) $\theta(g(n)) = \Omega(g(n))$ ($\theta = \text{big } \theta$)

(D) $\theta(g(n)) \cap \omega(g(n))$ is non-empty set. ($\theta = \text{small } \theta$)

Choose the correct answer from the options given below:

- (a) (A) only (b) (B) only
(c) (C) only (d) (D) only

Ans. (a) : (A) its worst - case running time is $O(g(n))$ and its best - case running time is

$\Omega(g(n))$. ($O = \text{big } O$) $\rightarrow \text{True}$.

Big - oh (O) denotes upper bound or worst case.

Big- omega (Ω) denotes lower bound or best case.

Big - Theta (θ) denotes Average bound or average case.

(B) its worst - case running time is $\Omega(g(n))$ and its best-case running time is $O(g(n))$. ($O = \text{big } O$)

false, worst case should be Big-oh and best case should be big - omega.

(C) $O(g(n)) = \Omega(g(n))$ ($O = \text{big } O$) False, cannot cover Big Omega.

(D) $\theta(g(n)) \cap \omega(g(n))$ is non-empty set ($\theta = \text{Small } \theta$)

false, no such $f(n)$ Exists, i.e. the inter section is indeed the Empty set.

53. Consider the following statements:

(A) Any tree is 2-colorable

(B) A graph G has no cycles of even length if it is bipartite.

(C) A graph G is 2-colorable if is bipartite.

(D) A graph G can be colored with $d+1$ color if d if the maximum degree of any vertex in the graph G .

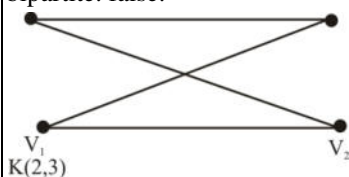
(E) A graph G can be colored with $O(\log |v|)$ colors if it has $O(|v|)$ edges.

Choose the correct answer from the options given below.

- (a) (C) and (E) incorrect
(b) (B) and (C) incorrect
(c) (B) and (E) incorrect
(d) (A) and (D) incorrect

Ans. (c): Any tree is 2- Colourable. True every tree is no loops and a bipartite graph with chromatic number 2. odd levels coloured with one colour and even level coloured with another colour.

• A graph G has no cycles of even length if it is bipartite. false.



• A graph G is 2- Colourable it is bipartite True. A graph $K(2,3)$ set of V_1 vertices coloured with one colour and set of V_2 vertices Coloured with another colour.

- A graph G can be colour with $d+1$ colours if d is the maximum degree of any vertex in the graph G . True as the vertex of degree d means it is adjacent to vertices Hence need a $d + 1$ different colours required.
- A graph G can be coloured with $O(\log|v|)$ Colours if it has $O|v|$ degrees. false as kn has $O(n^2)$ edges with Chromatic number n So it needs $O(\sqrt{v})$ Colours.

54. Which among the following statement is (are) true?

(A) A hash function takes a message of arbitrary length and generates a fixed length code.

(B) A hash function takes a message of fixed length and generates a code of variable length.

(C) A hash function may give same hash value for distinct messages.

Choose the correct answer from the options given below:

- (a) (A) only (b) (B) and (C) only
(c) (A) and (C) only (d) (B) only

Ans. (c): A Hash function H accepts a variable - length block of data M as input and produces a fixed - size hash value $h = H(M)$ In general terms, the principal object of a hash function is data integrity. A change to any bit or bits in m results, with high Probability in a change to the hash code.

55. Let G_1 and G_2 be arbitrary context free language and R an arbitrary regular language. Consider the following problems:

(A) Is $L(G_1) = L(G_1)$?

(B) Is $L(G_2) \leq L(G_1)$?

(C) Is $L(G_1) = R$?

Which of the problems are undecidable ?

Choose the correct answer from the options given below :

- (a) (A) only (b) (B) only
(c) (A) and (B) only (d) (A), (B) and (C)

Ans. (d) :

S r. n o.		FA Reg ular	PDA DCF L	P D A C F L	LBA CSL	Recu rsive HT M	RE LT M
1	Membership $W \in L$	D	D	D	D	D	UD
2	Finite ness- $L = \text{finite}$	D	D	D	UD	UD	UD
3	Equiv alence $L_1 = L_2$	D	D	U D	UD	UD	UD
4	Is L_1	D	UD	U	UD	UD	UD

•	$\subseteq L_2$ Subset checki ng			D			
5 •	Empti ness $L_1 = \phi$	D	D	D	UD	UD	UD
6 •	Disjoi nt operat or $L_1 \cap$ $L_2 = \phi$	D	UD	U D	UD	UD	UD
7 •	Is $L = \Sigma^*$ totalit y	D	UD	U D	UD	UD	UD
8 •	Is L Regul ar langua ge	D	D	U D	UD	UD	UD
* D = dicidable *UD = undecidable According to the above table, rows 3, 4 and 8 it's undecidable. Hence, the correct answer is (a), (b) and (c).							

56. Consider the following languages.

$$L_1 = \{ a^{\hat{Z}} \mid \hat{Z} \text{ is an integer} \}$$

$$L_2 = \{ a^{\hat{Z}} \mid \hat{Z} \geq 0 \}$$

$$L_3 = \{ \omega\omega \mid \omega \in \{a, b\}^* \}$$

Which of the language is (are) regular?

Choose the correct answer from the options given below:

- (a) L_1 and L_2 only
- (b) L_1 and L_3 only
- (c) L_1 only
- (d) L_2 only

Ans. (d) : Only the language L_2 is regular.

$$L_1 = \{ a^{\hat{Z}} \mid \hat{Z} \text{ is an integer} \}$$

$$\therefore L_1 = \{ a^1, a^4, a^{27}, a^{256}, \dots \}$$

$$L_2 = \{ a^{\hat{Z}} \mid \hat{Z} \geq 0 \}$$

$$\therefore L_2 = \{ \epsilon, a^{11}, a^{22}, a^{33}, \dots \}$$

$$L_3 = \{ \omega\omega \mid \omega \in \{a, b\}^* \}$$

$$L_3 = \text{set of all strings starting and ending with the same words.}$$

Explanation:-

L_1 does not contain any pattern to form a loop. Hence it's not possible to construct a DFA for L_1 .

- L_2 is a regular language since a DFA can be constructed using a loop that after every 11 symbols the automata reaches the final state.

in L_2 , once the word ' ω ' is scanned there is no way to compare the next word with the staring symbol of the first word.

57. Which of the following grammars in (are) ambiguous ?

(A) $s \rightarrow ss \mid asb \mid bsa \mid \lambda$

(B) $s \rightarrow asbs \mid bsas \mid \lambda$

(C) $s \rightarrow aAB$

$A \rightarrow bBb$

$B \rightarrow A \mid \lambda$ Where λ denotes empty string

Choose the correct answer from the options given below :

(a) (A) and (C) only

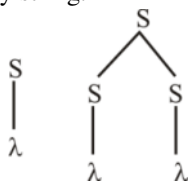
(b) (B) only

(c) (B) and (C) only

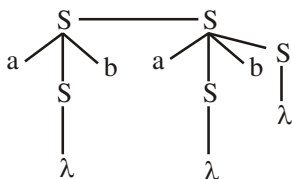
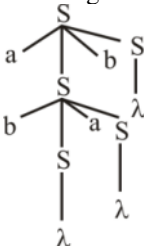
(d) (A) and (B) only

Ans. (d) :

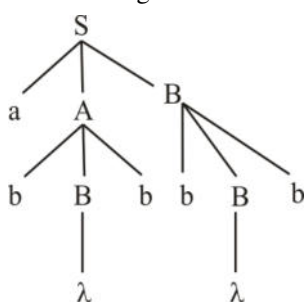
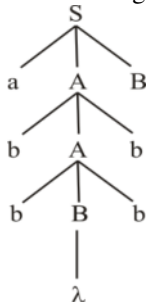
- A is ambiguous because λ can be generate using leftmost derivation having two different parse trees with an empty string.



- B is ambiguous grammar with string abab.



- C is an ambiguous grammar with string abbbb.



Hence the correct answer is (A) (B) and (C).

58. Which of the following cloud concepts is/are related to pooling and sharing of resources?

(A) Virtual Memory

(B) Service

(C) Virtualization

Choose the correct answer from the options given below:

(a) (C) only

(b) (A) and (B) only

(c) (A) only

(d) (B) only

Ans. (a) : Virtualization is a technique where multiple operating systems run simultanenously on a single machine virtually.

These machine can be accessed remotely by different users at the same time. Here, the same hardware like memory and CPU are utilized by all the operating systems in a systematic manner.

- 59. Post Office Protocol (POP) is a message access protocol which is used to extract message for client. In this regard, which of the following are true ?**
- (A) POP has two modes, Delete mode and keep**
 - (B) In Delete mode, mail is deleted from mail box after each retrieval.**
 - (C) In Delete mode, mail is deleted from mail box before each retrieval.**
 - (D) In keep mode, mail is deleted before retrieval.**
 - (E) In keep mode, mail remains in mail box after retrieval.**
- Choose the correct answer form the option given below:**
- (a) (A) and (B) only**
 - (b) (A), (D) and (E) only**
 - (c) (A), (B), (C) and (D) only**
 - (d) (A), (B) and (E) only**

Ans. (d) : Statement A :- TRUE-

Post office protocol (POP) is a type of computer networking and Internet standard protocol that has modes, delete mode and keep mode.

Statement -B (TRUE)-

In delete mode the mail is deleted from the mail box after each retrieval.

Statement E:- (TRUE) In keep made, mail remains in a mail box after retrieval.

Note:- Delete mode (delete after reading) keep mode (stores ever after reading).

- 60. Firewall is a device that filters access to the protected network from the outside network. Firewalls can filter the packets on the basis of**
- (A) Source IP Address**
 - (B) Destination IP Address**
 - (C) TCP Source port**
 - (D) UDP Source port**
 - (E) TCP Destination port**
- Choose the correct answer from the option given below:**
- (a) (A), (B) and (C) only**
 - (b) (B) and (E) only**
 - (c) (C) and (D) only**
 - (d) (A), (B), (C), (D) and (E) only**

Ans. (d) : Network layer firewall can filter on source and destination IP address source and Destination port no for both TCP and UDP packet.

• Firewall :- A firewall is a from of cyber security device that filters networks traffic. Network nodes can be separated from external traffic source, Internal traffic sources or even individual applications using firewall.

- 61. Consider the statements below.**
- A person who is radical (R) is electable (E) if he/she conservative (C), but otherwise is not electable.**

Few probable logical assertions of the above sentence are given below.

(A) $(R \wedge E) \Leftrightarrow C$

(B) $R \Rightarrow (E \Leftrightarrow C)$

(C) $R \Rightarrow ((C \Rightarrow E) \vee \neg E)$

(D) $(\neg R \vee \neg E) \vee C) \wedge (\neg R \vee \neg C \vee E)$

Which of the above logical assertions are true ?

Choose the correct answer form the options given below.

(a) (B) only

(b) (C) only

(c) (A) and (C) only

(d) (B) and (D) only

Ans. (d) :

1. $(R \wedge E) \Leftrightarrow C$ says that all (and only) conservatives are radical and electable. So this assertion is not true.

2. $R \Rightarrow (E \Leftrightarrow C)$ Say that same as the given assertion. this is a correct assertion.

3. $R \Rightarrow ((C \Rightarrow E) \vee \neg E) = \neg R \vee (\neg C \vee E \vee \neg E)$

Which if true for all interpretations. This is not a correct assertion.

4. $(\neg R \vee \neg E \vee C) \wedge (\neg R \vee \neg C \vee E) = (\neg R \vee (E \Rightarrow C)) \wedge (\neg R \vee (C \Rightarrow E)) = R \Rightarrow (E \Leftrightarrow C)$ Which is equivalent to assertion b.

This is also true.

62. Which of the following statements are true ?

(A) A sentence α entails another sentence β if β is true in few worlds where α is true.

(B) Forward chaining and backward chaining are very natural reasoning algorithm for knowledge bases in Horn form.

(C) Sound inference algorithm derive all sentences that are entailed.

(D) Propositional logic does not scale to environments of unbounded size.

Choose the correct answer from the options given below:

(a) (A) and (B) only

(b) (B) and (C) only

(c) (C) and (D) only

(d) (B) and (D) only

Ans. (d) :

• (B) For information bases in horn form true, forward and backward chaining is rather normal reasoning algorithms. these are very natural algorithm that run is linear time.

• (D) Propositional logic does not scale to unbounded conditions. That is right propositional logic is relatively efficient for certain tasks within an agent but it does not scale to unbounded environments because it lacks the descriptive capacity to deal concisely with time, space and universal patterns of object relationship.

Hence (b) and (d) correct options.

63. Which of the following statement are true ?

(A) Minimax search is breadth first, it processes all the nodes at a level before moving to a node in next level.

(B) The effectiveness of the alpha-beta pruning is highly dependent on the order in which the states are examined.

(C) The alpha-beta search algorithm computes the same optimal moves as minimax algorithm.

(D) Optimal play in game on imperfect does not reasoning about the current and future belief states of each players.

Choose the correct from the options given below:

- (a) (A) and (C) only (b) (A) and (D) only
(c) (B) and (C) only (d) (C) and (D) only

Ans. (c) : Statement (B)- The effectiveness of the alpha beta pruning is highly dependent on the order in which the states are examined.

Yes, it depends on which children are visited first. It's likely that no pruning happens if a node's children are visited in the worst possible order. We want to visit the best child first for max nodes so that time is not lost on the other children who are exploring worse scenarios we want to start with the worst child when it comes to min nodes.

Hence statement B is true.

Statement C:- The alpha-beta search algorithm computes the same optimal moves as the minimax algorithm.

True, it prunes nodes that have no bearing on the final outcome.

64. Match List with List II

Let $R_1 = \{(1, 1), (2, 2), (3, 3)\}$ and $R_2 = \{(1, 1), (1, 2), (1, 3), (1, 4)\}$

List I

List II

- | | |
|--------------------|--|
| (A) $R_1 \cup R_2$ | (I) $\{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (3, 3)\}$ |
| (B) $R_1 - R_2$ | (II) $\{(1, 1)\}$ |
| (C) $R_1 \cap R_2$ | (III) $\{(1, 2), (1, 3), (1, 4)\}$ |
| (D) $R_2 - R_1$ | (IV) $\{(2, 2), (3, 3)\}$ |

Choose the correct answer from the options given below:

- (a) A-I, B-II, C-IV, D-III
(b) A-I, B-IV, C-III, D-II
(c) A-I, B-III, C-II, D-IV
(d) A-I, B-IV, C-II, D-III

Ans. (d): $R_1 \cup R_2$ (union) is the relation containing all tuples that appear in R_1 , R_2 or both.

$$R_1 \cup R_2 = R_1 \cup R_2 = R_1 + R_2 - R_1 \cap R_2$$

$$R_1 \cup R_2 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (3, 3)\}$$

• $R_1 - R_2$ (Set difference)- is the relations containing all tuples of R_1 that do not appear in R_2

$$R_1 - R_2 = R_1 - R_1 \cap R_2$$

$$R_1 - R_2 = \{(2, 2), (3, 3)\}$$

• $R_1 \cap R_2$ (Intersection) - is the relation containing all tuples that appear in both R_1 and R_2

$$R_1 \cap R_2 = R_1 \cap R_2 = \{(1, 1)\}$$

• $R_2 - R_1$ (Set difference)- is the relation containing all tuples of R_2 that do not appear in R_1

$$R_2 - R_1 = R_2 - R_1 \cap R_2 = \{(1, 2), (1, 3), (1, 4)\}$$

Hence the correct answer is

$$A-I, B-IV, C-II, D-III$$

65. Match list I with List II

List I	List II
A. Handshaking	I. I/O interface inform the CPU that device is ready for transfer
B. Programmed I/O	II. requires two control signals working in opposite directions
C. Interrupt initiated I/O	III. has local memory & control large set of I/O devices
D. I/O processor	IV. require CPU to check the I/O flag & perform transfer

Choose the correct answer from the options given below:

- (a) A-I, B-II, C-III, D-IV
- (b) A-II, B-IV, C-III, D-I
- (c) A-II, B-IV, C-I, D-III
- (d) A-IV, B-III, C-II, D-I

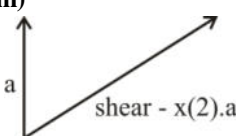
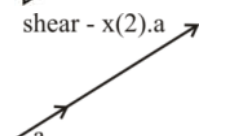
Ans. (c) : Hand shaking:- when one computer sends a message to another indicating that it wants to create a contact channel, it is called Handshaking. It requires two control signals working in opposite directions.

Programmed I/O:- The output of the I/O Instructions written in the computer programme is the trigger. It requires the CPU to check the I/O flag & perform the transfer.

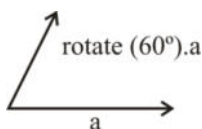
* Interrupt - Initiated I/O:- Interrupt powered I/O is a different way to deal with I/O. Interrupt I/O is a method of monitoring Input/output operation by sending a signal from a peripheral or terminal that needs to send or receive data. I/O interface informs the CPU that the device is ready for transfer.

* I/O processor:- A processor with direct memory access is know as an input output processor (IOP). The computer system is divided into memory units and processors in this way. I/O processor has local memory & control a large set of I/O devices.

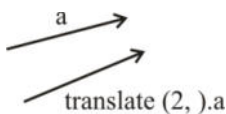
66. Match List I with List II, List gives 3×3 matrices representing 2D transformations and List-II shows the corresponding transformation diagrams.

List I (Matrix)	List II (Transformation Diagram)
A $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	(I) 
B $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$	(II) 

C $\begin{bmatrix} 0.5 & -0.866 & 0 \\ 0.866 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (III)



D $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (IV)



Choose the correct answer form the options give below:

- (a) A-IV, B-II, C-III, D-I
- (b) A-IV, B-III, C-II, D-I
- (c) A-III, B-II, C-IV, D-I
- (d) A-II, B-IV, C-III, D-I

Ans. (d) :

$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ it is the scaling matrix 2 in x and y-direction

by a factor.

Hence, A-II is the correct pair.

$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$ is the translation matrix vector 2 in each

direction x and y.

Hence B-IV is the correct pair.

$\begin{bmatrix} 0.5 & -0.866 & 0 \\ 0.866 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is an angle of 60 degrees rotation

matrix.

Hence C-III is the correct pair.

$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is the shear matrix of quantity 2 with only x-

shear.

Hence D-I is the correct pair.

67. Match List I with List II.

	List I		List II
(A)		(I)	Key attributes Type
(B)		(II)	Weak Entity Type
(C)		(III)	total Participation of Entity in a relation
(D)		(VI)	Multivalue Attribute Type

Choose the correct answer from the options given below:

- (a) A-II, B-IV, C-III, D-I
- (b) A-IV, B-I, C-II, D-III
- (c) A-II, B-I, C-IV, D-III
- (d) A-III, B-IV, C-I, D-II

Ans.(c):*E-R diagrams and their corresponding descriptions.

A-II, B-I, C-IV, D-III

68. Match List I with List II

With reference to CMM developed by Software Engineering Institute (SEI).

List I	List II
(A) INITIAL measurement	(I) Process
(B) REPEATABLE	(II) Process definition
(C) DEFINED	(III) Project management
(D) MANAGED	(IV) ADHOC

Choose the correct answer form the options given below:

- (a) A-III, B-IV, C-II, D-I
- (b) A-IV, B-III, C-I, D-II
- (c) A-IV, B-III, C-II, D-I
- (d) A-III, B-IV, C-I, D-II

Ans. (c) : Initial (Chaotic, adhoc, individual heroics):- the stage at which a new or undocumented repeat mechanism may be used.

Repeatable:- At the very least, the procedure is well recorded such that the same steps can be tried again.

Defined:- As a normal business procedure the process has been described and verified.

Managed (capable):- The method is Handled quantitatively according to a greed-upon metrics.

69. Match List I With List II

List I	List II
A. Branch and bound	I. Keeps track of all partial Paths which can be candidate for further exploration
B. Steepest accent hill climbing	II. Detects difference between current state and goal state
C. Constraint satisfaction	III. Discover problem state(s) that satisfy a set of constraints
D. Means end analysis	IV. Considers all moves from current state and selects best move

Choose the correct answer form the options given below:

- (a) A-I, B-IV, C-III, D-II
- (b) A-I, B-II, C-III, D-IV
- (c) A-II, B-I, C-III, D-IV
- (d) A-II, B-IV, C-III, D-I

Ans. (a): Branch-and-Bound:- Keeps track of all partial paths which can be a candidate for further exploration (Ex: A* search, BFS)

Steepest-accent hill Climbing:- Considers all moves from the current state and selects the best move (Ex. Hill-climbing algorithm)

Constraint satisfaction discovers problem state (s) that satisfy a set of constraints Ex (map coloring problem)

Means-end-analysis:- Detects the difference b/w the current state and the goal state Ex (means-end-analysis).

70. Match List I with List II

- | List I | List II |
|--|-----------------------|
| (A) Topological sort of DAG | (I) $O(V + E)$ |
| (B) Kruskal's MST algorithm | (II) $O(VE)$ |
| (C) Bellman Ford's single source shortest path algorithm | (III) $\theta(V + E)$ |
| (D) Floyd-Warshsall's all pair shortest path algorithm | (IV) $\theta(V^3)$ |

Choose the correct from the options below:

- (a) A-I, B-III, C-IV, D-II
- (b) A-III, B-I, C-IV, D-II
- (c) A-III, B-I, C-II, D-IV
- (d) A-I, B-III, C-II, D-IV

Ans. (d) : Topological sort of DAG = $O(V + E)$
Bellman - Ford's single - source shortest path algorithm = $O(VE)$

Kruskal's MST algorithm = $\theta(V + E)$ Due to sorting the edges the dominat term in kruskal time complexity is $O(E \log E)$.

Floyd-Warshall's all-pairs shortest path algorithm = $\theta(V^3)$.

71. Match List I with List II

L_R : Regular language, L_{CF} : Context free language

L_{REC} : Recursive language, L_{RE} : Recursively enumerable language.

- | List I | List II |
|-------------------------------------|----------------------------------|
| (A) Recursively Enumerable language | (I) $\bar{L}_{REC} \cup L_{RE}$ |
| (B) Recursive language | (II) $\bar{L}_{CF} \cup L_{REC}$ |
| (C) Context Free language | (III) $L_R^* \cap L_{CF}$ |

Choose the correct answer from the options given below:

- (a) A-II, B-III, C-I
- (b) A-III, B-I, C-II
- (c) A-I, B-II, C-III
- (d) A-II, B-I, C-III

Ans. (c) : $\bar{L}_{REC} \cup L_{RE} = (\text{Recursive} \cup \text{Recursively enumerable}) = \text{Recursively enumerable}$
Recursive languages closed under complement.
So, $\bar{L}_{REC}$ is Recursive union of recursive and recursively enumerable is recursively enumerable.
 $\bar{L}_{CF} \cup L_{REC} = (\text{Not context free} \cup \text{Recursive}) = (\text{Recursive} \cup \text{Recursive}) = \text{Recursive context free is not closed for a complement.}$
So, L_{CF} is Recursive. The union of recursive and recursive is recursive.
 $L_R \cap L_{CF} = (\text{Regular} \cap \text{Context free}) = \text{Context free.}$
The Intersections of regular and context free is context free.

72. Match List I with List II

List I		List II	
A.	Serial Line IP (SLIP)	I.	Application layer
B.	Border Gateway Protocol (BGP)	II.	Transport layer
C.	User Data Protocol (UDP)	III.	Data Link layer
D.	Simple Network management protocol	IV.	Network layer

Choose the correct answer form the options given below:

- (a) A-I, B-II, C-III, D-IV
- (b) A-III, B-IV, C-II, D-I
- (c) A-II, B-III, C-IV, D-I
- (d) A-III, B-I, C-IV, D-II

Ans. (b) : Serial line Internet protocol is a **Data link layer** Simple protocol that work with TCP/IP for communication over serial ports and routers.
Border Gateway Protocol is a Network layer used to exchange routing information for the internet and is the protocol used b/w ISP which are different ASes.
User data gram protocol is a Transport layer protocol. UDP is a part of the Internet protocol suite referred to as UDP/IP suite.
SNMP is an application layer protocol that uses UDP port number 161-162. SNMP is used to monitor the network, detect network faults and sometimes even used to configure remote devices.

73. Match List I with List II

List I		List II	
A.	Greedy Best First Search	I.	Space complexity is $O(d)$ where d =depth of the deepest optimal solution
B.	A*	II.	Incomplete even if the search space is finite

- | | |
|---------------------------------------|---|
| C. Recursive Best First Search | III. Optimal if optimal solution is reachable otherwise, returns the best reachable optimal solution |
| D. SMA* | IV. Computation and space complexity is too high |

Choose the correct answer from the option given below :

- (a) A-II, B-IV, C-I, D-III
- (b) A-III, B-II, C-IV, D-I
- (c) A-II, B-III, C-I, D-IV
- (d) A-III, B-IV, C-II, D-I

Ans. (a) :

- Greedy best first search algorithm always selects the path which appears best at that moment.
- A* is optimal and complete but time complexity and space complexity too high. it require more computation.
- Recursive best first search more memory knowledge but used only O(bd) memory.
- Memory simplified bound A⁺ such as A* optimal if you can jet optimal solution otherwise as it is memory bound.

74. Find the lexicographic ordering of the bit string given below based on the ordering 0 < 1.

- | | |
|-----------------|-----------------|
| (A) 001 | (B) 010 |
| (C) 011 | (D) 0001 |
| (E) 0001 | |

Choose the correct answer form the options given below:

- (a) 001 < 010 < 011 < 0001 < 0101
- (b) 0001 < 001 < 010 < 0101 < 011
- (c) 0001 < 0101 < 001 < 010 < 011
- (d) 001 < 010 < 0001 < 0101 < 011

Ans. (b) : Lexicographic order is nothing but dictionary order.

To determine the order of strings we need to compare each bit of all the strings one by one.

For Ex. Take 001 and 010, hence the second bits of the strings are 0, 1 respectively since 0 < 1 as per the given question 001 < 010

Similarly (0001 < 0001 < 010 < 0101 < 011)

75. Arrange the following types of machine is descending order of complexity:

- | | |
|-----------------|-----------------|
| (A) SISD | (B) MIMD |
| (C) SIMD | |

Choose the correct answer form the options given below:

- | | |
|-------------|-------------|
| (a) A, B, C | (b) C, B, A |
| (c) B, C, A | (d) C, A, B |

Ans. (c) :

- MIMD (Multiple Instructions Stream Multiple Data Stream)

This includes machines with multiple processing units that can apply multiple instruction to different data at the same time.

- SIMD (Single Instruction stream Multiple Data Stream)This means that a single Instruction is applied to several data sets at the same time.

- SISD (Single Instruction stream Single Data Stream)
This is the von Neumann principle of serial computer architecture, which only executes one instruction at a time.

76. Given below are different properties of 3D projections from A-D. Identify the correct order on the basis of property true of (i) a perspective projection only, (ii) an orthographic projection only, (iii) both orthographic and projective transformation and (iv) neither orthographic nor projective transformation respectively.

(A) Straight lines are mapped to straight lines.

(B) Distance and angles are (in general) preserved.

(C) Far away object appear the same size as closer ones.

(D) Requires homogeneous coordinates in order for it to be encoded into a linear transformation.

Choose the correct answer form the oprions given below:

(a) D, C, B, A

(b) B, C, D, A

(c) D, C,A, B

(d) C, D, B, A

Ans. (c) :

3D Projections	Properties
A perspective projection only	Requires homogeneous coordinates in order for it to be encoded into a linear transformation.
An orthographic projection only.	Far away object appear the same size as closer ones.
both orthographic and projective transformation.	Straight lines are mapped to straight lines.
neither orthographic nor projective transformation.	Distance and angles are (in general) preserved.

77. Arrange the following types of Cohesion form the to worst type.

(A) Logical Cohesion

(B) Sequential Cohesion

(C) Communication Cohesion

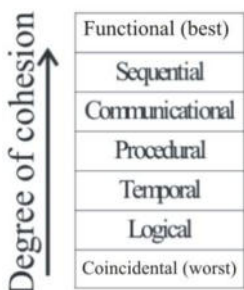
(D) Temporal Cohesion

(E) procedural Cohesion

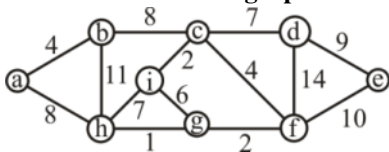
Choose the correct answer from the given below:

- (a) $A \rightarrow D \rightarrow E \rightarrow C \rightarrow B$
- (b) $A \rightarrow E \rightarrow D \rightarrow C \rightarrow B$
- (c) $B \rightarrow E \rightarrow C \rightarrow D \rightarrow A$
- (d) $B \rightarrow C \rightarrow E \rightarrow D \rightarrow A$

Ans. (d) :



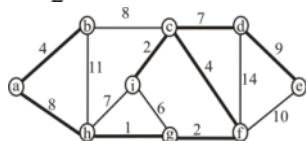
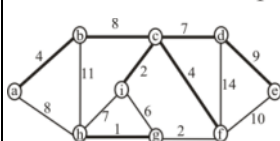
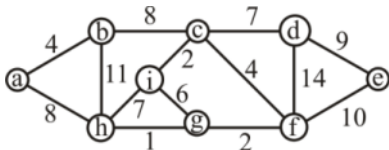
78. Consider the undirected graph below :



Using Prim's algorithm to construct a minimum spanning tree starting with node a, which one of the following sequences of edges represents a possible order in which the edges would be added to construct the minimum spanning tree?

- (a) (a, b), (a, h), (g, h), (f, g), (c, f), (c, i), (c, d), (d, e)
- (b) (a, b), (b, h), (g, h), (g, i), (c, i), (c, f), (c, d), (d, e)
- (c) (a, b), (b, c), (c, i), (c, f), (f, g), (g, h), (c, d), (d, e)
- (d) (a, b), (g, h), (g, f), (c, f), (c, i), (f, e), (b, c), (d, e)

Ans. (a & c): A minimum spanning tree (MST) or minimum weight spanning tree is a subset of the edge $(V - 1)$ of a connected, edge weighted undirected graph $G(V, E)$ that connects all the vertices together, without any cycle and with the minimum possible total edge weight.



So, there are two sequence (a, b), (b, c), (c, i), (c, f), (f, g), (g, h), (c, d), (d, e) with a cost 37. Another sequence is (a, b), (a, h), (g, h), (f, g), (c, f), (c, i), (c, d), (d, e) with a cost is 37.

79. Consider the following regular expressions:

(a) $r = a(b + a)^*$

(b) $s = a(a + b)^+$

(c) $t = aa^*b$

Choose the correct answer form the options given below based on the relation between the language generated by the regular expressions above:

(a) $L(r) \subseteq L(s) \subseteq (t)$ (b) $L(r) \supseteq L(s) \supseteq (t)$

(c) $L(r) \supseteq L(t) \supseteq (s)$ (d) $L(s) \supseteq L(t) \supseteq (r)$

Ans. (b) : $L(r) = \{a, aa, ab, aaa, aab, \dots\}$ This is a set of all strings starting with 'a'.

$L(s) = \{aa, ab, aaa, aab, \dots\}$ This is a set of all strings starting with 'aa' or 'ab'.

$L(t) = \{ab, aab, aaab, aaaab, \dots\}$ This is a set of all strings starting with 'a' ending with 'b' and contains any number of 'a's in between them.

$$L(r) \supseteq L(s) \supseteq (t)$$

Note :- Represent 0 or more number of symbols.

Represent 1 or more number of symbols.

80. Given below are two statements :

Statement I: 5 divides $n^5 - n$ whenever n is a nonnegative integer.

Statement II: 6 divides $n^3 - n$ wherever n is a nonnegative integer

In the light of the above statement, choose the correct answer form the option given below:

(a) Both statement I and statement II are correct

(b) Both statement I and statement II are incorrect

(c) Statement I is correct but statement II is incorrect

(d) Statement I is incorrect but statement II is correct

Ans. (a) : Statement I: 5 divides $n^5 - n$ Using mathematical Induction, if $n^5 - n$ is divide by 5, then $(n + 1)^5 - (n + 1)$ is also divisible by 5.

if 5 divides $n^5 - n = 5.k$

$$(n + 1)^5 - (n + 1) = n^5 + 2n^4 + 10n^3 + 10n^2 + 5n - n = 5.k + 5(n^4 + 2n^3 + 2n^2 + n)$$

This shows that $(n + 1)^5 - (n + 1)$ is a multiple of 5.

Hence it si always divisible by 5.

Statement II

$$n^3 - n = n(n^2 - 1)$$

$$= n(n + 1)(n - 1)$$

Hence n , $n + 1$ and $n - 1$ are three consecutive integers, one of them is a multiple of 2 and the other is a multiple of 3.

Hence 6 divides $n^3 - n$ is always divisible by 6 whenever n is a non- negative integer.

81. Given below are two statements:

Statement I: Hardwired control unit can be optimized to produce fast mode of operation.

Statement II: Indirect addressing mode needs two memory reference to fetch the operand.

In the light of the above statements, choose the correct answer from the options given below.

- (a) Both statement I and statement II are true
- (b) Both statement I and statement II are false
- (c) Statement I is correct but statement II is false
- (d) Statement I is incorrect but statement II is true

Ans. (a) : (I) Control signal are provided by a hard wired circuit in hardwired control techniques. The speed of a hardwired control unit can be optimized.

(II) Indirect addressing mode needs two memory references to fetch operand as the address of the operand is not given in instruction but the address field contains the address where the address of the operand is stored.

82. Given below are two statement:

Statement I: Bezier curves are curves that interpolate all the of their control points.

Statement II: A cubic bezier curve has four control points.

In the light of the above statements, choose the correct answer from the options given below.

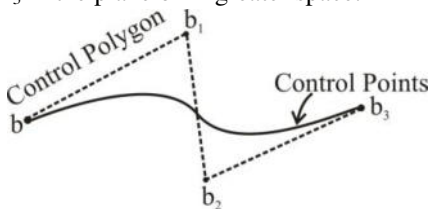
- (a) Both statement I and statement II are true
- (b) Both statement I and statement II are false
- (c) Statement I is correct but statement II is false
- (d) Statement I is incorrect but statement II is true

Ans. (d) : Statement I (incorrect)

The Bezier curve is defined by P_0 through P_n and n is named its order ($n = 1$ for the linear, 2 for the quadratic etc). The first and last checkpoints are often the end points of the curve, but interim check points do not usually lie on the curve.

Statement II (correct)-

The cubic Bezier curve is defined by four points P_0 , P_1 , P_2 and P_3 in the plane or in greater space.



Bezier Curve Example

A cubic Bezier curve has four control points.

83. Given below are two statements :

is two variables V_1 and V_2 are used for clustering then consider the following statement for k means clustering with $k = 3$:-

Statement I: If V_1 and V_2 have correlation of 1 the cluster centroid will be in straight line.

Statement II : If V_1 and V_2 have correlation of 0 the cluster centroid will be in straight line.

In the light of the above statements, choose the correct answer from the options given below.

- (a) Both statement I and statement II are true
- (b) Both statement I and statement II are false
- (c) Statement I is correct but statement II is false
- (d) Statement I is incorrect but statement II is true

Ans. (c) : Both data points would be in a straight line if the correlation between the variables V_1 and V_2 is 1. As a result all there cluster centroids will form a straight line

- 84. Assuming that the system call fork () never fails, consider the following C programs P1 and P2 executed on a UNIX VLinux system:**

<pre>/*P1*/ int main () { fork (); fork (); fork (); Printf("Happy\n"); }</pre>	<pre>/*P2*/ int main () { fork (); Printf("Happy\n"); fork (); Printf("Happy\n"); fork (); Printf("Happy\n"); }</pre>
---	---

Statement I: P1 displays "Happy" 8 times.

Statement II: P2 displays "Happy" 12 times.

In the light of the above statements, choose the correct answer from the options given below.

- (a) Both statement I and statement II are true
- (b) Both statement I and statement II are false
- (c) Statement I is correct but statement II is false
- (d) Statement I is incorrect but statement II is true

Ans. (c) : Fork () is a system call used to create a child process 'n' number of successive fork() call create $2^n - 1$ child processes.

In P₁:- Each child process prints one "Happy" parent process also prints one "Happy".

So, number of time P₂ displays "Happy" = $(2^3 - 1) + 1 = 8$.

In P₂:- At first fork () a new child process with two fork () and print f statements is created. Simailarly more child processes are created at each form () command

So, P₂ show "Happy" 14 (2 + 4 + 8) times

- 85. Given below are two statements:**

Statement I: Quality control involves the series of inspections, reviews and tests used throughout the software process, to ensure each work product meets the requirement placed upon it.

Statement II: Quality assurance consists of auditing and reporting functions of management.

In the light of the above statements, choose the correct answer from the options given below.

- (a) Both statement I and statement II are true
- (b) Both statement I and statement II are false
- (c) Statement I is correct but statement II is false
- (d) Statement I is incorrect but statement II is true

Ans. (a) : Quality control:- Quality control is a series of Inspections, reviews, and tests conducted during the software development process to ensure that each work product follows the specifications.

Hence statement I is correct.

$\Rightarrow$ **Quality assurance:-** Management auditing and monitoring activities makeup quality assurance. The aim of quality assurance is to provide management with the Information they need to make Informed decisions about product quality.

Hence statement II is correct.

86. Let G be a simple undirected graph, T_D be a DFS tree on G , and T_B be the BFD tree on G . Consider the following statements.

Statement I: No edge of G is a cross with respect to T_D .

Statement II: For every edge (u,v) of G , if u is at depth i and v is at depth j in T_B then $|i-j| = 1$.

In the light of the above statements, choose the correct answer from the options given below.

- (a) Both statement I and statement II are true
- (b) Both statement I and statement II are false
- (c) Statement I is correct but statement II is false
- (d) Statement I is incorrect but statement II is true

Ans. (c) : Statement-I Correct

Undirected graphs do not have cross edges in depth first search. But there can be cross edges in directed graphs.

Statement-II Incorrect

In Breadth first search an edge can be present b/w vertices of the same level. So $|i - j| = 0$.

87. Given below are two statements :

Statement I : The problem " $L_1 \wedge L_2 = \phi$?" is undecidable for context sensitive language L_1 and L_2 .

Statement II: The problem " $W \in L$?" is decidable for context sensitive language L . (where W is a string).

In the light of the above statements, choose the correct answer from the options given below.

- (a) Both statement I and statement II are true
- (b) Both statement I and statement II are false
- (c) Statement I is correct but statement II is false
- (d) Statement I is incorrect but statement II is true

Ans. (a) : The problem " $L_1 \wedge L_2 = \phi$?" is undecidable for context sensitive language L_1 and L_2 . Here $L_1 \wedge L_2$ is a disjoint operator it's undecidable problem.

Hence statement I is correct.

• The problem " $W \in L$?" is decidable for context sensitive language L (where w is a string). Here W is a string that is accepted by the particular language is a membership problem and it's decidable problem.

88. Given below are two statements :

Statement I: The laws of nature put two fundamental limits on data rate of channel. The H. Nyquist limit restricts the number of independent samples per second to twice the band width in a Noiseless channel.

Statement II: Shannon's major result about noised channel is that maximum data rate of a channel whose band width is H Hz, and whose signal to noise ratio is S/N is given by:

$$\text{Maximum number of bits/sec} = H \log_2 \left(1 + \frac{S}{N} \right)$$

In the light of the above statements, choose the correct answer from the options given below.

- (a) Both statement I and statement II are true
- (b) Both statement I and statement II are false
- (c) Statement I is correct but statement II is false
- (d) Statement I is incorrect but statement II is true

Ans. (a) : Noiseless channel :-

Nyquist bit rate for a noiseless channel the nyquist bit rate formula defines the theoretical maximum bit rate.

Bit rate = $2 \times$ band width $\times \log_2 L$

• Noisy Channel :- Shannon capacity in the reality. We cannot have a noiseless channel. the channel is always noisy.

Capacity = bandwidth $\times \log_2(1 + \text{SNR})$

In this formula, bandwidth is the bandwidth of the channel SNR is the signal to noise ratio and capacity is the capacity of the channel in bits per second.

89. Given below are two statements:

Statement I: In Ceaser Cipher each letter of Plain text is replaced by another letter for encryption.

Statement II: Diffie-Hellman algorithm is used for exchange of secret key.

In the light of the above statements, choose the correct answer from the options given below.

- (a) Both statement I and statement II are true
- (b) Both statement I and statement II are false
- (c) Statement I is correct but statement II is false
- (d) Statement I is incorrect but statement II is true

Ans. (a) : Caesar Cipher :- It is one of the simplest techniques for encryption where each letter of a given text is replace by a letter some fixed number of positions down the alphabet.

Ex- The text "ABC" can be encrypted using the ceasar cipher technique as "CDE" here the shift (Number of positions shifted) is 2.

Diffie-Hellman algorithm- Diffie Hellman algorithm is used for the exchange of secret key over an insecure channel.

• One the key is shared by both the sender and receiver, then the communication is done using the shared symmetric keys.

- 90. Given below are two statements:**
Statement I: A genetic algorithm is a stochastic hill climbing search in which a large populations of states is maintained.
Statement II: in nondeterministic environment agents can apply AND-OR search to generate contingent plans that reach the goal regardless of which outcomes occur during execution.
In the light of the above statements, choose the correct answer from the options given below.
- (a) Both statement I and statement II are true
 - (b) Both statement I and statement II are false
 - (c) Statement I is correct but statement II is false
 - (d) Statement I is incorrect but statement II is true

Ans. (a) : A genetic algorithm is to stochastic hill climbing algorithm that maintains a wide populations of states.
Hence statement I is correct.
Agents may use AND-OR search in non-deterministic environment to produce contingent plans that achieve the target regardless of which outcomes occur during execution.
Hence statement II Is correct.

- Question number (141-145)**
Question 91-95 concern a disk with a sector size of 12 bytes, 2000 tracks per surface 50 sectors per track, five double sides platters, and average seek time of 10 milliseconds.
- 91. If T is capacity of a track is bytes, and S is the capacity of each surface in bytes then, (T.S) = ____.**
- (a) (50 K, 50000 K)
 - (b) (25 K, 25000 K)
 - (c) (25 K, 50000 K)
 - (d) (40 K, 36000 K)

Ans. (c) : Track capacity = bytes/sector X sector/track = 512×50 = 25 K bytes/track
Surface capacity = bytes/track×track
surface = 25k×2000 = 50000 (bytes/...)

- 92. What is the capacity of the disk, in bytes?**
- (a) 25000 K
 - (b) 5000000 K
 - (c) 250000 K
 - (d) 50000 K

Ans. (b) : Capacity of the disk = bytes/sector x Sector/truck ×track / surface× surface/disk.
= 512×50 ×2000×5×2 = 5000000 K

- 93. Given below are tow statements :**
Statement I : The disk has a total number of 2000 cylinders.
Statement II: 51200 bytes in not a valid black size for the disk.
In the light of the above statements, choose the correct answer from the options given below.
- (a) Both statement I and statement II are true
 - (b) Both statement I and statement II are false
 - (c) Statement I is correct but statement II is false
 - (d) Statement I is incorrect but statement II is true

Ans. (a) : Statement-I The number of cylinder is 2000, which is the same or the number of tracks on each platter.

Statement-II The block size will be a multiple of the sector size. 51200 is not a valid block size in this case because block size cannot exceed the size of a track, which is 25600 bytes.

94. If the disk platters rotate at 5400 rpm (revolutions per minute), then approximately what is the maximum rotational delay?

- (a) 0.011 seconds (b) 0.11 seconds
(c) 0.0011 seconds (d) 1.1 seconds

Ans. (a) : In the disk platters rotate at 5400 rpm.
5400 rotational = 1 minute

$$\text{one rotation} = \frac{1}{5400} \times 60 \text{ seconds} = 0.011 \text{ sec}$$

95. If one track of data can be transferred per revolution, then what is the data transfer rate ?

- (a) 2850 KBytes/second
(b) 4500 KBytes/second
(c) 5700 KBytes/second
(d) 2250 KBytes/second

Ans. (d) : The capacity of a track is = bytes/sector x sector/track = 512x50= 25K

Data transfer rate = track size ÷ rotational day

If the disk platters rotate at 5400 rpm, the time required for one complete rotation will be,

$$\frac{1}{5400} \times 60 = 0.011 \text{ seconds}$$

Rotational delay = 0.011

Data transfer rate = 2250 K ÷ 0.011 = 220 (KBytes/sec)

Consider the following table structures related to a university for Q 96 to Q 100

<u>EMPLOYEE</u>		
NAME	VARCHAR (30)	NOT NULL
EID	VARCHAR (10)	NOT NULL
DEPTND	INT (5)	NOT NULL
HODEID	VARCHAR (10)	
SALARY	INT (10)	
PRIMARY KEY (EID).		
FOREIGN KEY (HODEID) REGRENCES EMPLOYEE (EID), FOREIGN KEY (DEPTNO) REGERNCES DEPARTMENT (DID):		
<u>DEPARTMENT</u>		
DID	INT (5)	NOT NULL
DNAME	VARCHAR (30)	NOT NULL
HODID	VARCHAR (10)	NOT NULL

HODNAME	VARCHAR (30)	
PRIMARY KEY (DID), UNIQUE (DNAME), FOREIGN KEY (HODID) REGERENCE EMPLOYEE (EID);		
<u>PROKECT WORK:</u>		
EMPIC	VARCHAR (10)	NOT NULL
PROGNO	INT (5)	NOT NULL
PROJECTLOC	VARCHAR (30)	NOT NULL
PRIMARY KEY (EMPID, RPOJNO), ROREIGN KEY (EMPIX)R\ REGERENCES EMPLOYEE (EID).		

96. On the basis of above given table structures, retrieve the distinct employee ID (EMPID) of all employee of university who are working in project No, 20, 30, and 40.

- (a) SELECT EMPID
FROM PROJECT WORK,
WHERE PROJNO = (20, 30, 40);
- (b) SELECT EMPID
FROM PROJECT WORK,
WHERE PROJNO IN (20, 30, 40);
- (c) SELECT DISTINCT EMPID
FROM PROJECT WORK,
WHERE PROJNO IN (20, 30, 40);
- (d) SELECT DISTINCT EMPID
FROM PROJECT WORK,
WHERE PROJNO = 20, 30, 40;

Ans. (c) : Since the primary key is a combination of two keys (EMPID, PROJNQ) so, duplicate values may occur in EMID and PROJNQ. So distinct key are required to eliminate duplicate in the combined table. AND IN is a set operator it matches all table values to the set values.

- SELECT DISTINCT EMPID
FROM PROJECT work
WHERE PROJNO IN (20, 30, 40).

97. Given below are two statement to find the sum of salaries of all employee of the English department as well as the maximum, minimum and average salary in English department.

STATEMENT I; SELCET SUM (SALARY),
MAX (SALARY),, MIN (SALARY), AVG
(SALARY) FROM EMPLOYEE,
DERPARTMENT WHERE DECTNO = DID
AND DNAME = 'ENGLISH'.

STATEMENT II: SELCET SUM (SALARY),
MAX (SALARY),, MIN (SALARY), AVG
(SALARY) FROM EMPLOYEE,
DERPARTMENT WHERE DNAME =
'ENGLISH'.

In the light of the above statements, choose the correct answer from the options given below.

- (a) Both statement I and statement II are true
- (b) Both statement I and statement II are false
- (c) Statement I is correct but statement II is false
- (d) Statement I is incorrect but statement II is true

Ans. (c): Two tables (join condition) must be matched so that records can be linked uniquely.

Query :- SUM (SALARY), MAX (SALARY), MIN (SALARY), AVG (SALARY) FROM EMPLOYEE, DEPARTMENT WHERE DEPTNO = DID AND DNAME = 'ENGLISH'

Without DEPTNO = DID condition as in statement II, there will be the Cartesian product of two tables (EMPLOYEE, DEPARTMENT) and it will appear many duplicate row with

DNAME = 'ENGLISH'

Hence statement I is correct but statement II is false.

98. Which of the following query/queries return the employee ID and name of employee whose salary is greater than the salary of all employee in department number 20 of university. Order result by employee ID (refer table structure given above).

(A) SELECT EID. NAME

FROM EMPLOYEE WHERE SALARY > (SELECT SALARY FROM EMPLOYEE WHERE DEPTNO = 20); ORDER BY EID

(B) SELECT EDI NAME

FROM EMPLOYEE WHERE SALARY > (SELECT SALARY FROM EMPLOYEE WHERE DEPTNO = 20);

(C) SELECT EID, NAME

FROM EMPLOYEE WHERE SALARY > ALL(SELECT FROM EMPLOYEE WHERE DEPTNO = 20) ORDER BY EID

Choose the correct answer from the options given below:

- (a) (A) and (B) only
- (b) (A) and (C) only
- (c) (B) only
- (d) (C) only

Ans. (d) : SELECT SALARY FROM EMPLOYEE WHERE DEPTNO = 20

The above inner query will be written the salary is of an employee whose department number is 20 this will match with the salary of the employee > salary of at least one employee.

Query :- SELECT EID, NAME

FROM EMPLOYEE

WHERE SALARY > ALL(SELECT FROM EMPLOYEE WHERE DEPTNO = 20)

ORDER BY EID

The query returns the employee ID and name of employees whose salary is greater than the salary of all employees in department number 20 of the university or order result by employee ID.

- 99. In reference to the above given table structure, which of the following query/queries will drop the 'SALARY' column from 'EMPLOYEE' table?**
- (A) **ABTER TABLE EMPLOYOEE DROP SALARY CASCASE.**
- (B) **ALTER TABLE EMPLOYEE DROP SALARY RESTRUCT.**
- (C) **ALTER EMPLOYEE DROP SALARY.**
- Choose the correct answer from the options given below:**
- (a) (A) and (B) only (b) (A) and (C) only
- (c) (B) and (C) only (d) (A) only

Ans. (a) : Alter table syntax:-

ALTER TABLE table-name

DROP [COLUMN] column-name [CASCADE | RESTRICT]

- The key word COLUMN, CASCADE, RESTRICT are optional. The default CASCADE. If you use the RESTRICT option, the column drop will be rejected if it would invalidate a dependent, schema object.
- Since the salary field is not present in any other table CASCADE | RESTRICT | has no impact.
- (c) ALTER EMPLOYEE DROP SALARY is incorrect syntax.

Hence the correct answer is (a) and (b) only.

- 100. Refer table structures given above University decided to give all employee in the 'SCIENCE' department a 20% rise in salary. Which of the following query/queries will compute the above results?**
- (A) **UPDATE EMPLOYEE
SET SALARY = SALARY*1.20
WHERE DEPT NO. IN (SELECT DID FROM DEPARTMENT WHERE DNAME = SCIENCE);**
- (B) **UPDATE TABLE EMPLOYEE
SET SALARY = SALARY*1.20
WHERE DBAME = SCIENCE;**
- (C) **ALTER TABLE EMPLOYEE
SET SALARAY = SALARY*1.20
WHERE DETINO, IN (SELECT DNAME FROM DEPRTMENT WHERE DNAME = 'SCICENCE')**
- Choose the correct answer from the options given below:**
- (a) (A) and (B) only
- (b) (A) only
- (c) (B) and (C) only
- (d) (C) only

Ans. (b) : Syntax :- UPDATE table _ name

SET column 1 = value 1, column 2 = value 2,
WHERE condition.

(A) UPDATE EMPLOYEE

SET SALARY = SALARY*1.20

WHERE DEPT NO. IN (SELECT DID

FROM DEPARTMENT WHERE DNAME = "SCIENCE");